A Hierarchical Control Approach For A Quadrotor Tail-Sitter VTOL UAV and Experimental Verification*

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***Abstract*— We present a hierarchical control approach that can be used to fulfill autonomous flight, including vertical takeoff, landing, hovering, transition, and level flight, of a quadrotor tail-sitter vertical takeoff and landing unmanned aerial vehicle (VTOL UAV). A unified attitude controller, together with a moment allocation scheme between elevons and motor differential thrust, is developed for all flight modes. A comparison study via real flight tests is performed to verify the effectiveness of using elevons in addition to motor differential thrust. With the well-designed switch scheme proposed in this paper, the aircraft can transit between different flight modes with negligible altitude drop or gain. Intensive flight tests have been performed to verify the effectiveness of the proposed control approach in both manual and fully autonomous flight mode.**

I. INTRODUCTION

In recent years, lightweight unmanned aerial vehicles (UAVs) have been widely used in various fields including aerial photography, facility detection, border patrol, disaster monitoring and relief, precision agriculture, parcel delivery, etc. Previously, people mainly applied either rotary-wing or fixed-wing aircraft to accomplish these tasks. However, in spite of its high maneuverability, a rotary-wing aircraft is inherently power inefficient by its flight mechanics, while a fixed-wing airplane, despite its high power efficiency, imposes rigorous demands for the takeoff and landing field. Because of these drawbacks, new UAV platforms are being actively sought to improve the efficiency and economy of flight related services. In particular, lightweight hybrid vertical takeoff and landing (VTOL) UAVs, which can hover at a stationary point and take off or land vertically, like a rotary-wing aircraft, but at the same time cover a long distance, like a fixed-wing aircraft, have attracted immense recent interests [1]–[3].

Among the various kinds of VTOL UAVs (namely, tilt-rotor, tilt-wing, rotor-wing, dual-systems, and tail-sitter) [4], a tail-sitter UAV has the simplest mechanical structure, which in turn reduces manufacturing complexity and saves considerable weight. Through changing the vehicle's attitude, a tail-sitter VTOL UAV can transit between hovering flight and level flight, thus achieving both maneuverability and efficiency. The price to be paid is the complicated dynamics and increased difficulty in controller design, especially during transition and hovering flight. During the transition flight, the vehicle will experience a wide range of angles

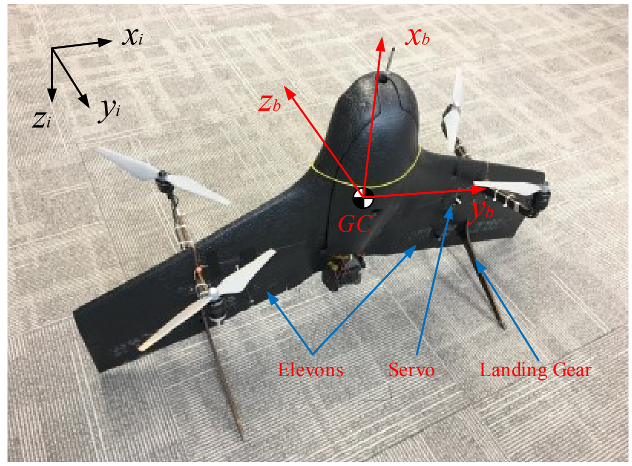

Fig. 1. The quadrotor tail-sitter (Coordinate systems are defined in Section II), Video available at: `https://youtu.be/dVE9Q76AALs`

of attack, causing the airflow to separate and producing a nuisance aerodynamic turbulence. Cross wind can cause strong disturbance on the wing and degrade the hovering accuracy and landing safety. Other challenges include the aerodynamic coupling between motors and wings, high angle of attack wing aerodynamics and propeller aerodynamics. Due to these obstacles, flight controller design to achieve fully autonomous operation, precise landing and hovering, as well as trajectory tracking, of a tail-sitter VTOL UAV is still a big challenge.

So far, several prototypes and implementation methods have been proposed in the literature. Oosedo *et al.* [5] proposed two types of tail-sitter UAVs (namely, cross-type and asterisk-type), which are composed of a flying wing without any control surface, but with four motors for attitude and thrust control. The authors conducted a comparison study, showing that the cross type has poor performance due to the interaction between the propeller slipstream and the vehicle wing during hovering. As a result, they proceeded with the asterisk-type and investigated its transition, which is achieved by sending a step pitching command to a standard quadrotor controller used for hovering control of the UAV. Later, in [6], Oosedo *et al.* focused on solving the problem of the transition process and proposed two optimal transition strategies from hovering to level flight (namely, minimizing the transition time and minimizing transition time with constant altitude) and performed experimental verification. Hochstenbach *et al.* [7] also proposed a similar configuration (namely, asterisk-type tail-sitter composed of a flying wing without a control surface, but with four rotors) for autonomous parcel delivery. A middle-level control mode,

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which gives a smooth decreasing pitch command during the transition process, is added to the quadrotor mode and plane mode flight controllers. Experiments on the transition process and turning flight at a high angle of attack were performed. Later, Theys *et al.* [8] proposed a velocity controller by continuously varying the transition angle and thrust to reach a prescribed velocity based on an asterisk-type tail-sitter UAV without a controller surface.

Unlike the tail-sitter vehicles mentioned above, which mainly depend on differential thrust to gain enough moment to control the attitude, Bapst *et al.* [9] developed a platform with a flying wing and two rotors. They proposed a cascaded controller which receives the desired velocity and calculates the required attitude and thrust for the entire flight envelope. Due to the fact that the airspeed sensor being used can only measure airspeed precisely at small angle of attack, the controller was only verified at level flight conditions by experiments. In [10], Verling *et al.* developed another similar platform and proposed a full attitude controller. Using a high-level navigation loop, the vehicle can hover, transit to level flight, and loiter around a specified point with a single attitude controller.

Our target is to design and implement a lightweight VTOL UAV which can achieve flight, including takeoff, hovering, level flight and landing, autonomously in a real outdoor environment where modest wind is occasionally present. In our previous work [11], we designed and implemented a quadrotor tail-sitter prototype which consists of four rotors and a flying wing without a control surface. A simple PID position controller and a nonlinear attitude controller were used to verify the performance of our vehicle. As shown by experiments, the implemented prototype has a negligible vibration level, robust landing capacity, and acceptable hovering accuracy in the presence of cross wind. The hovering accuracy is within 25.4 cm with 5.2 m/s wind. Outdoor experiments were conducted to show that the designed vehicle can successfully transit between hovering flight and level flight.

This paper, continuing to improve on the reported UAV prototype in [11], addresses flight controller design problems. We feature a versatile, hierarchical flight control system that contains two operation modes: manual mode and autonomous mode. The manual mode provides a reliable attitude and altitude controller for the quadrotor tail-sitter, during hovering flight, transition flight, and level flight. This enables easy human-in-the-loop operation. The effectiveness of using elevons to enhance attitude control is investigated in this mode. For the autonomous mode, all the flight modes (namely, rotary-wing mode, transition mode, and fixed-wing mode) and transitions among them are fully autonomous. The operator only needs to supply high-level commands, such as way points and transition commands, ahead of real flight. A fully autonomous mission is conducted to verify the effectiveness of our control system.

The remainder of this paper is organized as follows. In Section II, we briefly introduce our vehicle system. In Section III, the overall controller architecture and controller design are presented. In Section IV, flight tests in manual mode are conducted to verify the flight controller as well as the effectiveness of using elevons. After that, fully autonomous missions are experimentally verified. In Section V, conclusions are drawn and future works are presented.

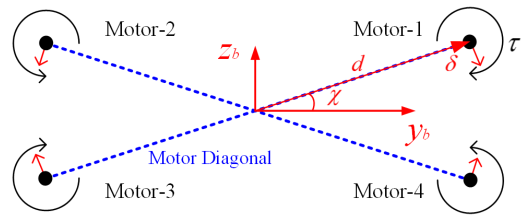

Fig. 2. The quadrotor rotors layout

II. System Configuration

A. Aircraft Design

As shown in Fig. 1, the quadrotor tail-sitter consists of a flying wing and four rotors. The motors are inclined along the motor diagonal axis by $\delta = 15°$ (shown in Fig. 2) which can considerably increase the rotation moment along x_b and hence increase the vehicle's stability in the presence of cross wind. The landing gears are designed such that the vehicle can successfully land with a maximal tilt angle of $35°$, the associated cross wind that causes such tilt angle is 6.5m/s based on wind tunnel test data. And the location of motor support, which transmits the vibration from the motor and propeller to the main body, is carefully calculated and assembled to attenuate the overall vibration of our vehicle [11]. Two elevons are added to improve the flight performance based on previous design.

B. Coordinates System

The local north-east-down (NED) coordinate system is chosen as the inertial frame. Body coordinates used in this paper are the same as those of conventional fixed-wing aircraft. Rotation along x_b, y_b and z_b are defined as roll, pitch, and yaw, respectively. Z-X-Y Tait-Bryan angles, direct cosine matrix, and quaternion are used to represent the UAV attitude in different situations. Detailed explanation of this can be found in [12].

C. Avionics

The core avionics of our vehicle includes a Pixhawk Auto-pilot, a GPS module, and a pitot tube airspeed meter. Using the Pixhawk open source software stack [13], all sensor measurements are fused via an extended Kalman filter to estimate the vehicle position, velocity, attitude, angular velocity, airspeed, etc. A pair of 915-MHz telemetry radios are used to communicate between the flying vehicle and a ground station. The ground station is mainly used to monitor the state of our vehicle and to set way points and other types of commands before flight [14].

III. Hierarchical Control System Design

There are three flight modes in our system: rotary-wing mode, fixed-wing mode, and transition mode, and two levels of operation modes: manual mode and autonomous mode.

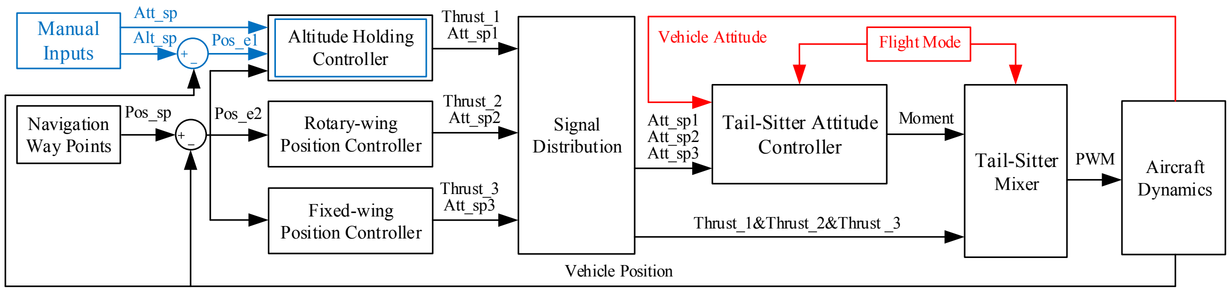

Fig. 3. Architecture of the hierarchical control system

The architecture of our flight control system is shown in Fig. 3. It is seen that the whole flight system consists of three levels: position controller, attitude controller, and mixer.

There are three position controllers in the first level: altitude holding controller, rotary-wing position controller, and fixed-wing position controller, which are running simultaneously. A position controller receives the desired position and computes the desired attitude and thrust, which are then fed to a signal distribution module. Then, the signal distribution module sends all desired attitudes, which come from the three position controllers, to a tail-sitter attitude controller. Similarly, the thrusts of all three position controllers are fed to a tail-sitter mixer for further processing. The second level is the tail-sitter attitude controller. Depending on the current vehicle flight mode, the attitude will choose the proper reference attitude and compute the desired moment. The third level is the tail-sitter mixer, which is used to receive the moment from the attitude controller and choose the proper thrust from the three position controllers according to the current flight mode. And then, by using a proper moment allocation scheme, as presented in Section III-D, the tail-sitter mixer maps the moment and thrust to the actuators' movements according to the current flight mode.

In autonomous mode, the vehicle, if in rotary-wing mode and receiving a transition command from the ground station, will execute the forward transition from rotary-wing mode to fixed-wing mode by calling the altitude holding controller. During this process (i.e., in transition mode), a linearly decreasing pitch angle command is fed to the altitude holding controller, while the roll angle is set to zero, and the yaw angle is set to be the same as the current vehicle yaw value. Once the prescribed pitch angle and airspeed are both reached, the fixed-wing mode will be triggered. Similarly, the vehicle in fixed-wing mode will execute the backward transition from fixed-wing mode to rotary-wing mode by calling the same altitude holding controller if a transition command is given. During this process (i.e., also in transition mode), a linearly increasing pitch angle is fed to the altitude holding controller. Other settings are similar to the forward transition. Similarly, the rotary-wing mode will be triggered if another prescribed airspeed and pitch angle are both reached.

In manual mode, only the altitude holding controller in the first level is running. This operation mode is particularly useful for aircraft performance testing as it effectively stabilizes the aircraft. In manual mode, the pilot can directly control the roll and pitch angles during level flight and the full attitude during hovering flight. Both manual and autonomous modes share the same control architecture. In the following parts, we will explain our contributions in detail.

A. Position Controller

The fixed-wing position controller mainly comes from [15], while the rotary-wing position controller comes from [16]. The two position controllers have been implemented on the existing framework of the Pixhawk Auto-pilot software stack [13]. We directly adopt these two mature modules to our system. Here we focus on introducing our Altitude Holding Controller" module (as shown in Fig. 3), which plays crucial role in transition maneuvers. In this module, an altitude controller with aerodynamic feedforward (described in Section III-B) is developed to hold the vehicle altitude to form a stable system that is suited for human-in-the-loop operation. The rigid body altitude dynamics can be written as

$$\begin{aligned} \ddot{h}_z &= \frac{1}{m} r_3^T \left(F_{aero} + F_T\right) + g \\ R &= \begin{bmatrix} r_1 & r_2 & r_3 \end{bmatrix} \end{aligned} \tag{1}$$

where h_z is the altitude in the inertial frame, with its positive direction along the Z axis of the inertial frame, $F_{aero} \in \mathbb{R}^3$ and $F_T = \begin{bmatrix} F_{Tx} & F_{Ty} & F_{Tz} \end{bmatrix}^T \in \mathbb{R}^3$ are the aerodynamic force and motor thrust vector in the body frame, respectively, and $R \in SO(3)$ is the rotation matrix from the body frame to inertial frame. The reference altitude is described as h_d, and the altitude error is described as $h_e = h_d - h_z$. A simple altitude PID controller is used to calculate motor thrust as follows:

$$r_3^T \left(F_{aero} + F_T\right) + mg = k_p h_e + k_i \int h_e dt + k_d \frac{dh_e}{dt} \tag{2}$$

The resulting thrust can be finally described as

$$F_{Tx} = \frac{k_p h_e + k_i \int h_e dt + k_d \frac{dh_e}{dt} - mg - r_3^T F_{aero}}{r_{31}} \tag{3}$$

where r_{31} is the first entry of $r_3 \in \mathbb{R}^3$.

B. Aerodynamic Feedforward

The aerodynamic feedforward term is mainly in the longitudinal direction and can be written as

$$F_{aero} = \begin{bmatrix} -\cos(\alpha) & 0 & \sin(\alpha) \\ 0 & 1 & 0 \\ -\sin(\alpha) & 0 & -\cos(\alpha) \end{bmatrix} \begin{bmatrix} D \\ 0 \\ L \end{bmatrix} \tag{4}$$

where D and L are respectively the drag and lift force represented in the stability frame [17] and α is the angle of attack. The aerodynamic force can be parameterized as

$$D = \frac{1}{2}\rho V^2 S C_D(\alpha), \quad L = \frac{1}{2}\rho V^2 S C_L(\alpha) \tag{5}$$

where ρ is the air density, S is the reference area of the vehicle, and $C_D(\alpha)$ and $C_L(\alpha)$ are respectively the drag and lift coefficients, which are measured via a full-scale wind tunnel experiment [12]. The test results and fitted lines are shown in Fig. 4. The airspeed V and angle of attack α are defined as

$$V = \|u\|, \qquad \alpha = \arctan\left(\frac{u_z}{u_x}\right) \tag{6}$$

where $u = \begin{bmatrix} u_x & u_y & u_z \end{bmatrix}^T$ is the airspeed vector represented in the body frame.

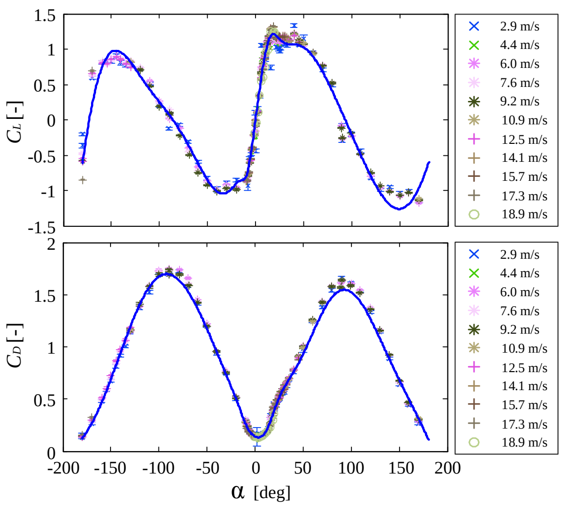

Fig. 4. Aerodynamic coefficients

C. Attitude Controller

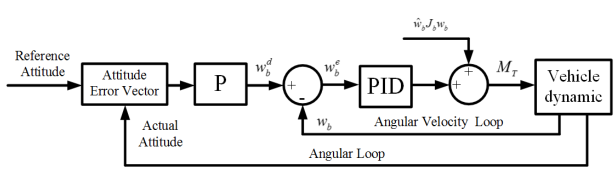

Fig. 5. Attitude controller structure

For the attitude controller, we adopt a cascaded control structure, as shown in Fig. 5. The outer loop is an angular loop and the inner loop is an angular velocity loop. According to Bullo *et al.* [18], the outer loop, which is a proportional controller, will be exponentially stable if the angular velocity can be controlled to the desired one. We compare different attitude error decomposition methods and show them in Fig. 6. $\|e_{att}\|$ is the magnitude of the attitude error vector. Because of symmetry, $\|e_{att}\|$ for the rotations along the roll, pitch, or yaw axis is identical. Different methods are labeled using the different legend names shown in Fig. 6. "LEE" is from [19]. We can see that $\|e_{att}\|$ is essentially a sine function, a nonlinear function with respect to the rotation

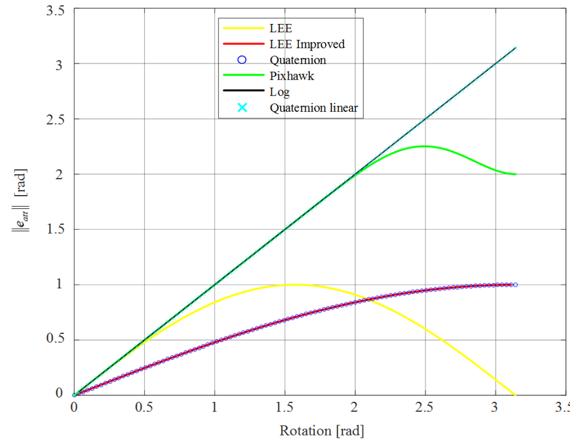

Fig. 6. Mag. of att. error of single axis rotation

angle. Hence, the controller convergence rate decreases at large initial error. The author further proposed an improved attitude error decomposition method in [20], denoted as "LEE Improved" in Fig. 6, to improve the convergence rate. Unfortunately, this approach is essentially the same as "Quaternion" [21] as shown in Fig. 6.

The good degree of linearity between the actual rotation error and the decomposed attitude error (i.e. e_{att}) can guarantee a uniform and good convergence rate for all initial attitude errors [20]. The "Pixhawk" [13] method (V1.6.0) is a combination of [21] and [22]. We can see from Fig. 6 that it becomes nonlinear as the rotation radian becomes larger. The "Log" method [11] can guarantee the linearity. But it has singularity when $|\varphi| = \pi$. We propose a "Quaternion linear" method, which computes the rotation axis and angle directly from quaternion, thus eliminating the singularity of conventional log map.

$$\begin{aligned} q_e &= {q_d}^* \otimes q = \begin{bmatrix} q_{e0} & q_{e1} & q_{e2} & q_{e3} \end{bmatrix}^T \\ e_{att} &= \frac{\varphi}{\sin\frac{\varphi}{2}} \cdot \begin{bmatrix} q_{e1} & q_{e2} & q_{e3} \end{bmatrix}^T \end{aligned} \tag{7}$$

where q_d, q, and q_e are respectively the reference attitude, actual attitude, and attitude error represented in quaternion. $(\cdot)^*$ is the conjugate of a quaternion, and $\otimes$ is the quaternion product [23]. The first entry of the quaternion is the scalar and the remaining three parameters are the vector part.

After getting the attitude error vector, the reference angular velocity in the body frame can be written as

$$w_b^d = -k_p e_{att}, k_p > 0 \tag{8}$$

The inner loop is a PID controller, and a feedforward term is added to cancel the Coriolis term. The control algorithm can be written as

$$M_T = K_p w_b^e + K_i \int w_b^e dt + K_d \frac{dw_b^e}{dt} + \hat{w}_b J_b w_b \tag{9}$$

where $M_T = \begin{bmatrix} M_{Tx} & M_{Ty} & M_{Tz} \end{bmatrix}^T \in \mathbb{R}^3$ is the moment vector, K_p, K_i, and K_d are positive diagonal gains for the PID controller, $w_b^e = w_b^d - w_b$ is the angular velocity error.

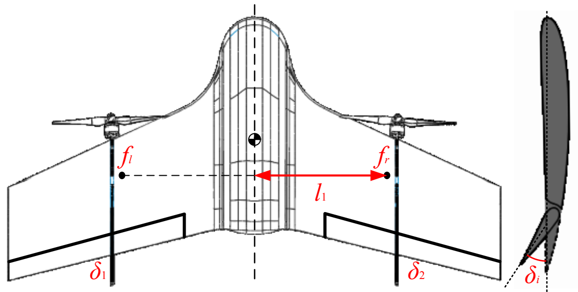

Fig. 7. Vehicle configuration and nomenclatures of elevons

D. Tail-Sitter Mixer

After getting the trust and moment, a tail-sitter mixer is used to allocate them to the proper actuators. As mentioned in Section II-A, we further incorporate a pair of elevons to enhance the stability for level and transitional flight. The configuration of our vehicle is shown in Fig. 7. δ_i is the elevon deflection angle. f_l and f_r are respectively the lift forces produced by the left wing and right wing, and l_1 is the distance between the lift forces, which acts on the mean aerodynamic chord (MAC), and longitudinal axis. According to [24], f_l and f_r can be represented as

$$\begin{aligned} f_l &= \frac{1}{2}\rho V^2 S_l \left(C_{L_0} + C_{L_\alpha}\alpha + C_{L_\delta}\delta_1\right) \\ f_r &= \frac{1}{2}\rho V^2 S_r \left(C_{L_0} + C_{L_\alpha}\alpha + C_{L_\delta}\delta_2\right) \end{aligned} \tag{10}$$

where S_l and S_r are respectively the left and right wing area. Because of symmetry, S_l and S_r are the same. C_{L_0} is the lift coefficient when $\alpha = 0$ and $\delta_i = 0$, C_{L_α} and C_{L_δ} are respectively two lift coefficient terms derived with respect to the angle of attack and elevons deflection. Similarly, the aerodynamic pitch moment can be written as

$$\begin{aligned} M_l &= \frac{1}{2}\rho V^2 S_l \left(C_{m_0} + C_{m_\alpha}\alpha + C_{m_\delta}\delta_1\right)\bar{c} \\ M_r &= \frac{1}{2}\rho V^2 S_r \left(C_{m_0} + C_{m_\alpha}\alpha + C_{m_\delta}\delta_2\right)\bar{c} \end{aligned} \tag{11}$$

where $\bar{c}$ is the mean aerodynamic chord, C_{m_0} is the pitch moment coefficient when $\alpha = 0$ and $\delta_i = 0$, and C_{m_α} and C_{m_δ} are respectively two pitch moment coefficient terms derived with respect to the angle of attack and elevon deflection. Finally, the elevon angles can be determined by

$$\begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} \frac{2(1-a_1(V))M_{Tx}}{\rho V^2 S_l l_1 C_{L_\delta}} \\ \frac{2(1-a_2(V))M_{Ty}}{\rho V^2 S_l \bar{c} C_{m_\delta}} \end{bmatrix} \tag{12}$$

Due to the motor inclination in our design, the four rotor thrusts can be determined by

$$\begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} \frac{F_{Tx}}{\cos(\delta)} \\ \frac{a_1(V)M_{Tx}}{\sin(\delta)d+\kappa\cos(\delta)} \\ \frac{a_2(V)M_{Ty}}{(\cos(\delta)d-\kappa\sin(\delta))\sin(\chi)} \\ \frac{M_{Tz}}{(\cos(\delta)d-\kappa\sin(\delta))\cos(\chi)} \end{bmatrix}$$

$$T_i = c_t \varpi_i^2, \qquad \kappa = \frac{c_q}{c_t} \tag{13}$$

where d is the length of each motor from the aircraft center of gravity (shown in Fig. 2), χ is the included angle between the motor diagonal and y_b, T_i is the rotor thrust, ϖ_i is the rotation speed of each rotor, c_t and c_q are the propeller thrust coefficient and propeller torque coefficient respectively, κ is the ratio between the torque coefficient and thrust coefficient. $a_1(V)$ and $a_2(V)$ are two functions related to airspeed to weight the contribution of the moment from elevons and rotors in roll and pitch direction, which are computed as

$$a_i(V) = \begin{cases} 1 \\ 1 - \left(k_{i,1}\frac{V}{V_T} + k_{i,2}\right), (i = 1, 2) \\ 1 - (k_{i,1} + k_{i,2}) \end{cases} \tag{14}$$

where $a_i(V) \in \left[1 - (k_{i,1} + k_{i,2}),\ \ 1\right]$, $k_{i,1}$ and $k_{i,2}$ are two constants for roll direction $(i = 1)$ and pitch direction $(i = 2)$. In our implementation, $k_{1,1} = 0.7$, $k_{1,2} = 0.15$, $k_{2,1} = 0.6$, $k_{2,2} = 0.2$. V_T is a barrier speed that the aircraft has to exceed in order to switch to full fixed-wing mode in autonomous mode. In rotary-wing mode, transition mode, and fixed-wing mode, $a_i(V)$ equals 1, $1 - \left(k_{i,1}\frac{V}{V_T} + k_{i,2}\right)$, and $1 - (k_{i,1} + k_{i,2})$, respectively.

IV. Experimental Verification

In this section, we present real outdoor flight results to validate the proposed control system. The outdoor experiments are divided into two parts. One is flight tests with and without elevons in manual mode, and the other is fully autonomous flight. During the outdoor experiment, the reported wind speed is around 4 m/s.

A. Verifying the Effectiveness of Elevons

We conducted a comparison study between the UAV attitude control with and without elevons. The flight test results of the vehicle without elevons are shown in Fig. 8 and Fig. 9. At first, the vehicle was in hovering flight. Then, a command was given by the pilot to transit to level flight at 83.32s. After that, a roll command was given to initiate a bank turn. While we can see a well-controlled attitude response during hovering and transition, maneuvers (e.g., bank turn starting at 84.27s) during level flight cause an oscillatory attitude response, as seen in Fig. 8. Correspondingly, motor saturations are observed, as shown in Fig. 9. It can be concluded that, due to the increased aerodynamic moment but decreased motor thrust at high-speed level flight, attitude control relying solely on the four motors can barely provide sufficient moment for stable attitude control. The motors are operating near saturation, resulting in an oscillatory response in the attitude loop. The possible causes are set out as below:

- Effect of advanced ratio: Rotating at the same RPM, the thrust and torque produced by a typical propeller will decrease with the increase of advanced ratio, which is linearly dependent on the airspeed along x_b [25]. As a result, the motors need to rotate faster to provide the same control moments when flying forward, which easily leads to motor saturation.

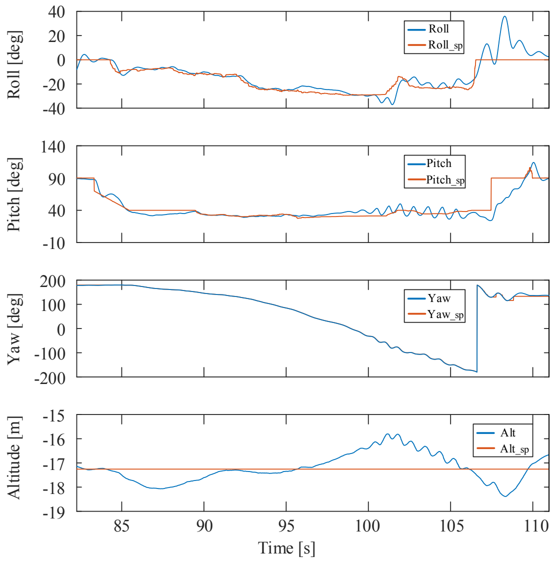

Fig. 8. Attitude and altitude without elevons

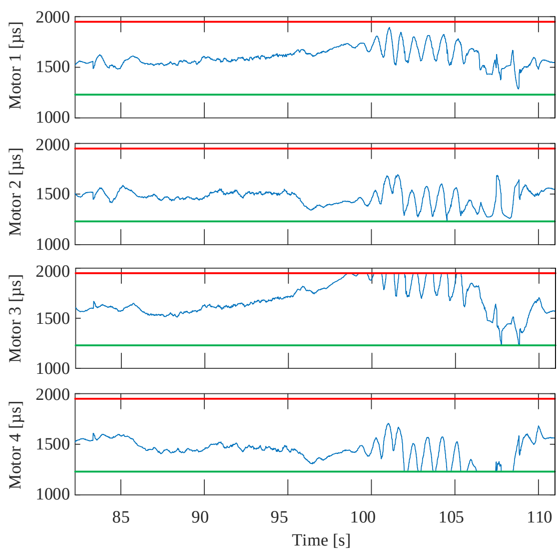

Fig. 9. Motor PWM without elevons

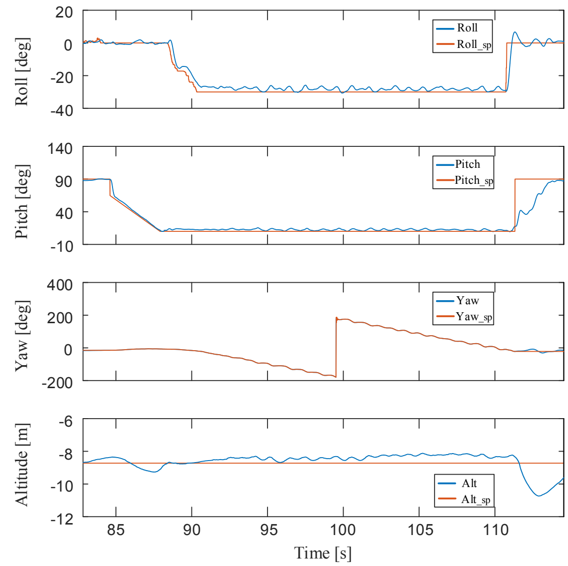

Fig. 10. Attitude and altitude with elevons

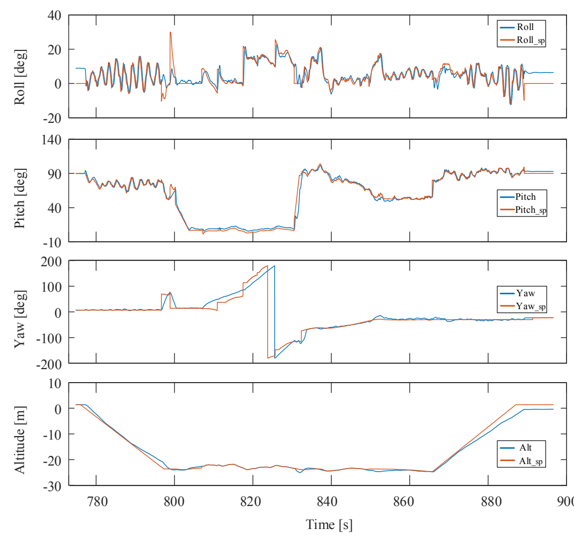

Fig. 11. Attitude and altitude in autonomous flight

- Effect of wing sweeping: As indicated in [24], a swept wing has a similar effect to that of a wing dihedral and can enhance the roll stability in level flight. In other words, extra rolling moment is required to keep the vehicle at none-zero rolling angle. As shown in Fig. 9, the PWM of Motor 1 and Motor 3 are generally larger than Motor 2 and Motor 4 to provide this rolling moment, thus adding extra loading to motor saturation.
- Effect of angle of attack: The PWM of Motor 1 and Motor 2 is overall lower than that of Motor 3 and Motor 4, which means the vehicle tends to pitch upward during the bank turn. Since our main airframe is adopted from a commercial off-the-shelf flying wing, which is primarily designed as static stable in longitudinal direction, the aerodynamic moment of the main airframe tends to adjust its angle of attack to the cruise angle of attack if any deviation occurs. The four motors therefore needs to provide a necessary moment to overcome such aerodynamic moment to maintain the desired pitch angle.

Considering the above possible causes, there are a number of ways to mitigate the above problems:

- Eliminate the wing sweeping design to reduce its stability in roll direction.
- Choose propellers with higher pitch angle to provide increased control moments.
- Use motors with higher KV such that higher RPM can be achieved.
- Add elevons to provide additional moments.

In this paper, we adopt the last method, which is shown to be very effective and easy to implement. The flight test results with elevons are shown in Fig. 10. It can be seen that the attitude and altitude can be well tracked during the entire flight, including hovering, transition (84.6s-87.9s and 111.3s-114.4s), stable level flight (87.9s-111.3s), and maneuvers during level flight (88.4s-111.3s).

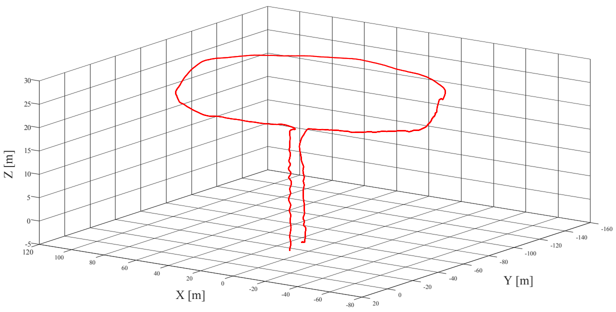

Fig. 12. Position in autonomous flight

B. Autonomous Flight

The autonomous flight test results are shown in Fig. 11 and Fig. 12. A ground station is used to set navigation way points and plan takeoff, transition, and landing commands. The autonomous flight procedure is as follows. The vehicle first takes off and flies to a prescribed altitude. After the altitude is reached, the vehicle flies in rotary-wing mode to a specified way point to prepare for forward transition flight. After the vehicle reaches the way point with its z_b pointing to the next way point. The vehicle begins to pitch forward and transits to fixed-wing mode. The vehicle flies through some navigation way points in fixed-wing mode and, then transits back to rotary-wing mode at a prescribed way point. After that, the vehicle flies to above the landing point and lands, thus completing the whole mission. The entire process mentioned above was performed fully autonomously. The maximum altitude drop and gain are respectively 0.67 m and 1.2 m during the transition process. We executed the same autonomous mission six times, and all were successful.

V. Conclusion and Future Work

We have proposed a hierarchical control approach that can be operated in both autonomous mode and manual mode. In autonomous mode, a flight mission, including takeoff, hovering, transition, level flying, and landing were accomplished autonomously. In manual mode, a comparison study via real flight tests was performed to verify the effectiveness of using elevons, in addition to motor differential thrust. A unified attitude controller, together with a stable moment allocation scheme between the elevons and rotors, was developed for different flight modes. With the well-designed mode switch scheme, we can transit between different modes with a maximum altitude drop of 0.67 m and gain of 1.2 m. Intensive flight tests indicate that the proposed control system is very reliable and consistent. Future work will focus on online system identification and model-based robust controller design using systematic controller analysis and synthesis tools to characterize and improve the flight performance.

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